Experiment 13

1. Aim

To study and analyze the limitations of low-code development platforms using Zoho Creator and the Library Book Issue Management application, focusing on customization, performance, scalability, integration, security, cost, and platform dependency, and to identify possible solutions for overcoming these limitations.

2. Limitations of Low-Code Development Platforms

2.1 Limited Customization

Low-code platforms provide predefined layouts, components, and templates that simplify application development. However, they may restrict developers who require complete control over the user interface, design, and application behavior. Highly customized layouts and advanced user experiences may require additional scripting or external development tools.

Solution: Use custom pages, supported UI components, and scripting features wherever available.

2.2 Performance Limitations

Application performance may decrease as the number of records, workflows, reports, and calculations increases. Complex scripts, repeated database queries, and unnecessary processing can increase execution time and reduce responsiveness.

Solution: Optimize Deluge scripts, filter records before processing, retrieve only required data, and minimize unnecessary workflow executions.

2.3 Scalability Challenges

Scalability refers to an application's ability to handle increasing numbers of users, records, and transactions. Low-code platforms may have limits on data storage, workflow execution, API calls, and concurrent usage.

Solution: Review platform limits, optimize application design, monitor usage, and consider higher-capacity plans or external services when necessary.

2.4 Integration Limitations

Low-code platforms may not directly support every external service, database, or legacy system. Integrations can require API configuration, authentication, custom scripting, or third-party services, increasing development complexity.

Solution: Use supported APIs, webhooks, connectors, and integration services while ensuring secure data exchange.

2.5 Platform Dependency (Vendor Lock-in)

Applications built on Zoho Creator may rely on platform-specific features, configurations, and Deluge scripts. Migrating to another platform can require significant effort because application components and business logic may not transfer directly.

Solution: Maintain technical documentation, back up important data, and evaluate migration requirements before selecting a platform.

2.6 Cost and Usage Limitations

Low-code platforms may offer different subscription plans with varying limits on users, storage, automation, integrations, and advanced features. As application usage grows, additional resources or higher subscription plans may be required, increasing operational expenses.

Solution: Review subscription limits, estimate future requirements, and choose a suitable plan based on expected usage and budget.

2.7 Security and Compliance Challenges

Although low-code platforms provide security features, developers must configure permissions and access controls correctly. Improper configurations can expose confidential information or allow unauthorized users to access or modify records. Organizations must also consider applicable data-protection requirements.

Solution: Implement role-based access control, restrict access to confidential fields, validate inputs, and review audit logs regularly.

2.8 Limited Offline Access

Cloud-based low-code applications generally depend on internet connectivity for many of their functions. Offline capabilities may be limited, depending on the platform and application configuration. Network interruptions can affect access to records, form submissions, and workflow execution.

Solution: Evaluate available offline capabilities and establish alternative procedures for temporary connectivity issues.

2.9 Limitations in Advanced Technologies

Low-code platforms support various automation and AI capabilities, but highly specialized algorithms, advanced machine-learning models, and computationally intensive operations may require additional development. Integrating such technologies can increase complexity and dependence on external services.

Solution: Use supported integrations and external AI services where appropriate, while considering security, performance, and maintenance requirements.

Summary: Limitations and Solutions

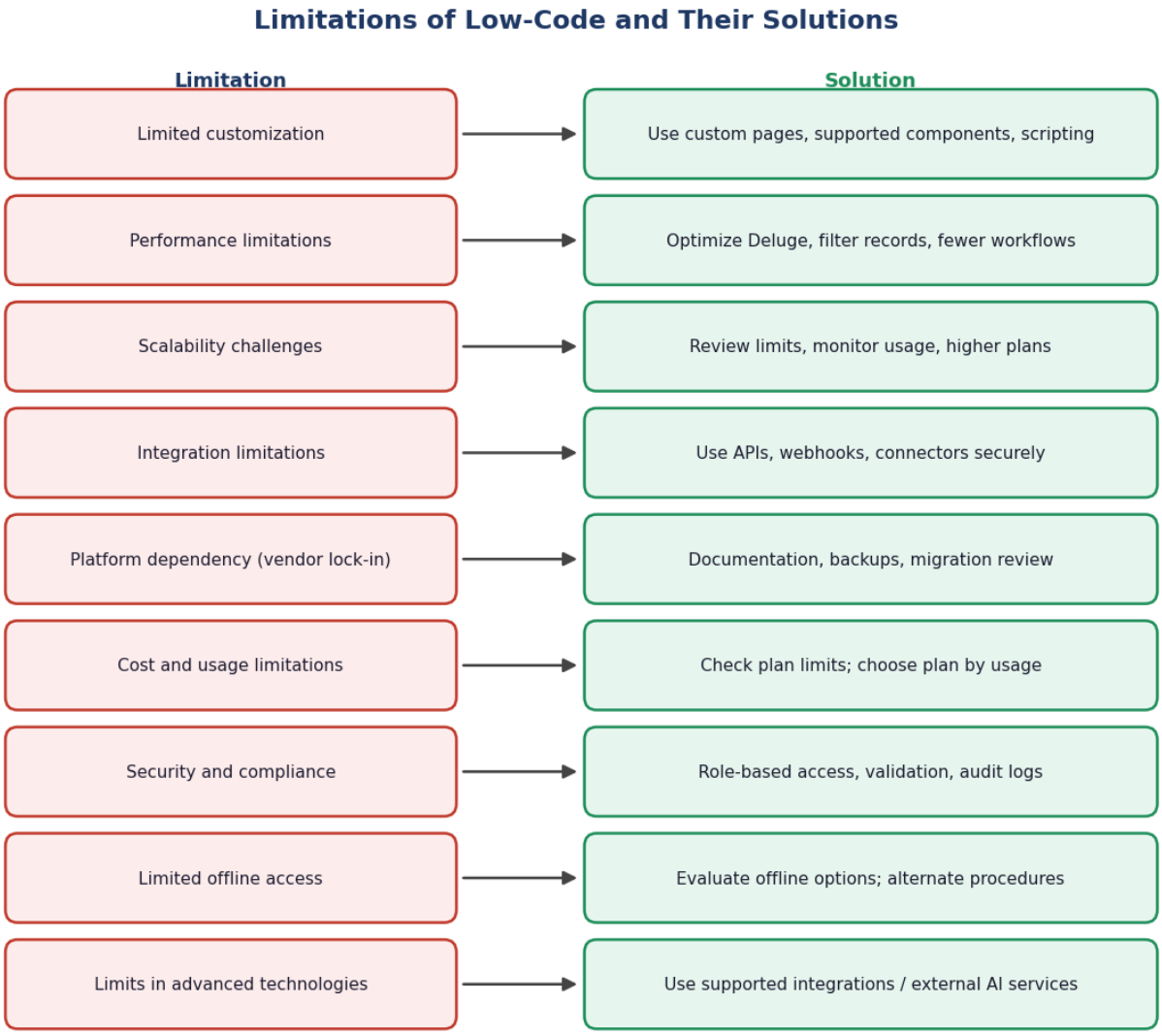


Figure 13: Limitations of low-code and their solutions

Conclusion

Low-code development platforms such as Zoho Creator simplify application development by reducing coding effort, accelerating development, and providing built-in tools for forms, reports, workflows, and dashboards. However, these platforms have limitations related to customization, performance, scalability, integration, platform dependency, cost, security, offline access, and advanced technologies. These limitations must be considered during application planning and implementation.